AI/ML-Based Predictive Maintenance and Vehicle Health Monitoring System

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***Abstract*—Predictive maintenance plays an important role in improving vehicle reliability, reducing unexpected failures, and minimizing maintenance costs. This paper presents an AI based predictive maintenance and vehicle health monitoring system for identifying abnormal vehicle conditions and predicting potential component failures. The proposed system uses vehicle parameters including engine temperature, RPM, oil pressure, vibration, battery voltage, coolant temperature, fuel consumption, vehicle speed, and operating hours. The sensor data is preprocessed and analyzed using machine learning techniques to estimate failure probability and vehicle health. The system generates a vehicle health score, failure probability, maintenance status, and information about the major contributing factors. Explainable artificial intelligence techniques are incorporated to improve the interpretability of model predictions. The proposed approach demonstrates the potential of machine learning for condition-based vehicle monitoring and early maintenance alerts.**

Keywords— Predictive Maintenance, Vehicle Health Monitoring, Machine Learning, Random Forest, XGBoost, Explainable AI, Vehicle Failure Prediction

I. INTRODUCTION

Modern Vehicles use multiple sensors to continuously monitor operating conditions such as temperature, RPM, oil pressure, vibration, and battery voltage. Analysing this sensor data using Artificial Intelligence and Machine Learning can help detect abnormal conditions and predict possible vehicle failures before they occur. Predictive maintenance can therefore reduce unexpected breakdowns, maintenance costs, and vehicle downtime.

II. PROBLEM STATEMENT

Unexpected vehicle and component failures can affect vehicle reliability and safety. Traditional maintenance methods may not identify developing faults early. This project addresses this problem by using vehicle sensor data and machine learning to estimate failure probability and monitor overall vehicle health.

III. OBJECTIVES

The main objectives of the proposed system are:

1. To analyse important vehicle operating parameters.
2. To preprocess sensor data for machine learning.
3. To predict possible vehicle or component failures.
4. To calculate a vehicle health score and failure probability.
5. To identify important factors contributing to abnormal conditions.
6. To generate maintenance alerts based on vehicle condition.

IV. RELATED WORK

Machine Learning techniques have been increasingly used for predictive maintenance and fault detection. Classification algorithms such as Decision Tree, Random Forest, and XGBoost can identify patterns associated with abnormal operating conditions. Deep learning models such as LSTM and GRU can also be used for sequential sensor data. Explainable Artificial Intelligence can further help interpret model predictions by identifying important contributing parameters.

V. PROPOSED SYSTEM

The proposed system monitors vehicle sensor parameters and processes the collected data using machine learning techniques. The model estimates

failure probability and generates a vehicle health score. Based on the predicted condition, the system provides a maintenance status such as NORMAL, WARNING, or CRITICAL.

A. *SYSTEM ARCHITECTURE:* The system follows the sequence:

Vehicle Sensors -> Data Preprocessing -> ML Model -> Failure Probability -> Vehicle Health Score -> Maintenance Alert.

B. *INPUT PARAMETERS:* The system considers engine temperature, RPM, oil pressure, vibration, battery voltage, coolant temperature, fuel consumption, vehicle speed, and operating hours.

C. *VEHICLE HEALTH MONITORING:* The predicted vehicle condition is represented using a health score and failure probability. These values are used to classify the vehicle condition and provide an appropriate maintenance alert.

VI. METHODOLOGY

The sensor data is first prepared and preprocessed before being provided to the machine learning model. Relevant vehicle parameters are used as input features. Machine Learning algorithms are then trained to identify patterns related to vehicle condition and potential failure.

A. *DATA PREPROCESSING: The dataset is checked and prepared for model training. The input features are separated from the target variable, and the data is divided into training and testing sets.*

B. *MACHINE LEARNING MODEL:* Machine Learning algorithms such as- Random Forest and XGBoost are used for predictive analysis. The trained model predicts the likelihood of vehicle failure based on the input sensor parameters.

C. *VEHICLE HEALTH SCORE:* The model prediction is converted into a vehicle helth score representing the overall operating condition of the vehicle. A higher health score indicates a healthier vehicle condition.

D. *EXPLAINABLE AI:* Feature importance is used to identify the vehicle parameters that contribute most to the model prediction. This helps explain why a vehicle may be classified as NORMAL, WARNING, or CRITICAL.

VII. RESULTS AND DISCUSSION

The developed system was evaluated using the test data to predict vehicle condition based on sensor parameters. The model provides failure probability and vehicle health status, while feature importance identifies the parameters that most influence the prediction. Based on the predicted condition, the system generates NORMAL, WARNING, or CRITICAL maintenance alerts.

VIII. CONCLUSION AND FUTURE WORK

The proposed AI-based predictive maintenance system uses vehicle sensor data to monitor vehicle health and predict possible failures. It provides failure probability, health status, and maintenance alerts. Future work can include real-time sensor integration, larger datasets, cloud connectivity, and advanced machine learning models.

IX. REFERENCES

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